LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety precautions to be observed while working on compass, inertial navigation systems and safety precaution while working with nonmagnetic
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety precautions to be observed while working on compass, inertial navigation systems and safety precaution while working with nonmagnetic tools on compass.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of non-magnetic tool usage and compass protection; Instructor supervision required for all handling of sensitive equipment.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft magnetic compass?
(b) Why must non-magnetic tools be used near the compass?
(c) What are the basic components of an Inertial Navigation System (INS)?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Magnetic interference sources • Compass protection zones • Non-magnetic tool identification • Handling procedures for compass • INS safety protocols	L & Ds	PPT, IB	15 min
(b)	• Demonstration of non-magnetic tools • Simulation of compass handling • Review of EMI/EMC risks • Safety checklist verification • Instructor supervision techniques	L & Ds	PPT, PDF	20 min
(c)	• Summary of safety rules • Q&A on specific scenarios • Reinforcement of key points • Preparation for next lesson • Final safety briefing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three specific safety precautions for working near the aircraft compass.
(b) Explain why standard steel tools are prohibited in the INS area.
(c) Describe the immediate action required if a magnetic tool is dropped near the compass.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between magnetic and inertial navigation systems, referencing the block diagram to be studied in LP 10.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Simple inertial navigation system block diagram.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min